

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 1st Term Examination, 2020
HUM3117
(Accounting and Auditing for Planners)

Full Marks: 120

Time: 1.5hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Answer any **two** questions from each section in separate script.

Section – A

1. (a) What is account? Show the classification of account according to modern method. (15)
- (b) What is transaction? Describe the characteristics of transaction. (15)

2. Mr. Ifaz started a business on January 2020. The transactions are as follows:

2020	
January -1	Investment made in business by Mr. Ifaz cash Tk. 1,50,000
January -2	Office rent paid in cash Tk. 10,000
January -3	Cash deposited into bank Tk. 50,000
January -4	Purchased supplies on account Tk. 7,000
January -5	Received Tk. 25,000 from a customer in payment for future delivery services.
January -6	Performed delivery services for customer for cash Tk. 20,000
January -7	Paid the Tk. 7,000 account payable (transaction January 4)
January -8	Billed customers for delivery services performed Tk. 30,000
January -9	Received a bill for advertising that appeared in a local newspaper Tk. 1,500
January -10	Advertisement bill paid by cheque (transaction January-9)

Required:

- (a) Prepare journal (20)
 - (b) Prepare any four ledger accounts (10)
3. (a) Define trial balance. What are the objectives of preparing trial balance? (10)
 - (b) From the following ledger balances of Tahmid Company, prepare a trial balance as on December 31, 2020: (20)

Account titles	Balance (tk.)
Cash	50,000
Accounts Receivables	12,000
Notes payable	11,000
Unearned revenues	3,500
Owners' equity	35,000
Withdrawals	1,500
Commission received	18,000
Furniture	22,000
Accumulated depreciation	11,000
Sales	25,000
Rent paid	8,000
Salary expense	3,000
Utility expense	7,000

Section – B

4. (a) What is audit methodology? In which stage of auditing these methodologies are employed? (15)
(b) Distinguish between external and internal audit. (15)
5. (a) Define audit process. State the importance of planning the audit procedure on a prior basis. (15)
(b) What is audit risk? What are the means of exposing audit risk? (15)
6. (a) What is meant by tax? Describe the objectives of taxation in modern public finance. (20)
(b) Differentiate between direct and indirect taxes with examples. (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rdYear 1stTerm Examination, 2020

URP 3121
(Rural Development Planning-I)

Full Marks: 120

Time: 1.5 hrs

- N.B.**
- i) Answer any Two questions from each section in separate scripts.
 - ii) Figures in the right margin indicate full marks.
 - iii) Assume data if necessary

Section – A

1. (a) “Rural development is assessed by using certain indicators”-elaborate. Current pandemic situation of the country has significantly affected the indicators of rural development-how? Explain using necessary example. (15)
- (b) “Relationship between rural branding strategy and local economic development actions is vital for effective rural development”-justify with example. (15)
2. (a) Demarcation of rural development using GDP calculation has certain limitations-explain. (05)
- (b) Suppose, a society consists of 5 families whose incomes are \$ 8300, \$14000, \$ 25000, \$ 46000 and \$ 90000. Construct a Lorenz curve for this five family economy with quintile calculation. (10)
- (c) How is poverty line measured with indicators? Illustrate with an example. (15)
Distinguish between absolute and relative measurement of rural poverty.
3. (a) Narrate the concept and significance of “Poverty Gap” in rural development planning. Make a summery on the advantages and disadvantages of Poverty Gap Index (PGI). (15)
- (b) “Elaborate the concept and indicators used in Physical Quality of Life Index (PQLI) for poverty gap identification. Calculate PQLI for the following countries and explain the result. (15)

Country	Life expectancy at age one	Infant mortality	Literacy
	Year	Per 1000 live birth	Percent
A	65	165	25
B	71	41	36
C	66	83	85

Section – B

4. (a) What are the general considerations for defining rural area? Make a brief overview on parameters that distinguishes urban and rural areas in respect of Bangladesh. Use necessary example. (15)
- (b) Identify and explore major factors influencing rural settlement pattern with appropriate example. Use necessary sketches. (10)
- (c) Discuss the significance of spatial planning for rural areas of Bangladesh. (05)
5. (a) Critically analyze V-AID programme in rural development of Bangladesh. (15)
- (b) Make a summery on significant rural development programs including their success and failures in Bangladesh. (15)
6. (a) Define threshold population and its significance in rural planning. Provide a brief overview on threshold population calculation for any service. (15)
- (b) Narrate the concept of “Digital Village”. Use of digital technology for rural service-facilities planning is vital for Bangladesh especially considering current pandemic situation-explain. (15)

Khulna University of Engineering & Technology
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BURP 3rd Year 1st Term Examination, 2020

URP 3161
(Housing and Real Estate Development)

Full Marks: 120

Time: 1.5 hrs

- N.B.** i) Answer any **two** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Please write down the fundamental issues of defining Formal/ Informal Housing. (20)
Which type of housing is quite easy to adopt as per your understanding?
Provide an interlinked diagram to prove your thought.
(b) What is the difference between 'need' and 'want'? With this continuation which (10)
is the best answer for student's Residential Hall in KUET?
2. (a) One continuing problem in Bangladesh is "affordability" in housing sector. What (10)
will be your suggestion for future housing policy to meet up the challenges?
(b) Make comparison table for the following using necessary example. (20)
i) Between Apartments and Condominiums
ii) Between Land Ceiling and Land Banking
3. (a) Suppose you are assigned to calculate the tax of urban housing property. What (10)
will be your considerations? According to Wendt's model calculate the land
value if Net income = 100000 and CR = 350?
(b) How many occurrences are mainly responsible for urbanization process in (10)
Indonesia? Please briefly discuss about the occurrences.
(c) Today's world is for the richer people. How can you relate it with the terms (10)
"insurgence" and "informality"?

Section – B

4. (a) True real estate decisions are about acquiring, financing, using, improving and (15)
disposing of actual real estate assets. Explain the statement.
(b) Tangible assets are different from intangible assets. How? Explain with relevant (15)
examples in context of real estate.
5. (a) Why investment in renovation and urban development is highly risky? Explain in (15)
terms of cost and cash flow uncertainty. Also explain the mitigation strategies to
overcome the uncertainty.
(b) When will the downturn actually occur in a real estate market? Explain with (15)
supply and demand curve. How a developer will survive during a market
downturn?
6. (a) What is real estate valuation? Summarize the primary principles of real estate (15)
valuation with an example.
(b) In the hypothetical situation of simple bid rent theory, consider there are 318 (15)
households. Among them there are 26 doctors, and 292 other commuters of
which 80 walk and 212 drive. Calculate and draw the bid rent curve of each
group. Assume necessary data for each group.

Khulna University of Engineering & Technology
Department of Urban and Regional Planning

BURP Regular Examination, 2020

CE 3107

(Elements of Solid Mechanics)

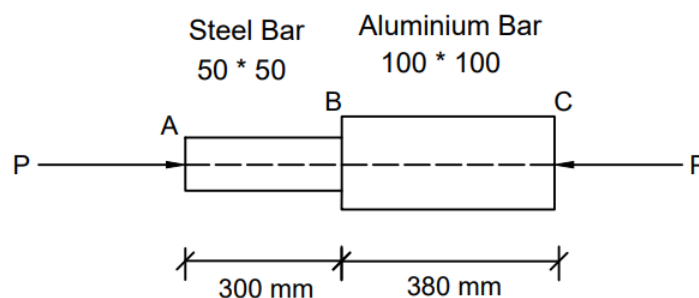
Full Marks: 120

Time: 1.50 hrs

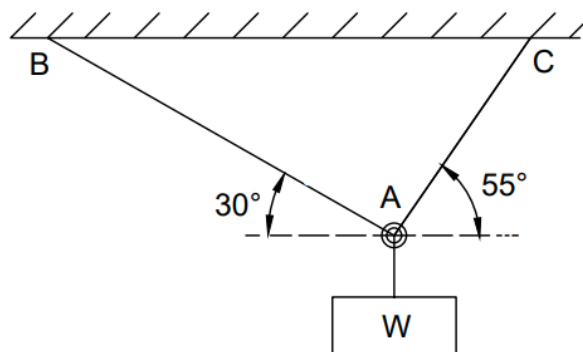
- N.B.**
- i) Answer any two questions from each section in separate script.
 - ii) Figures in the right margin indicate full marks.
 - iii) Assume reasonable value if necessary

Section – A

1. (a) Define stress, strain and elasticity. Derive a relationship between stress and strain of an elastic body. (08)
- (b) A member formed by connecting a steel bar to an aluminium bar is shown in the following figure. Assume that the bars are prevented from buckling sidewise, calculate the magnitude of force P , that will cause the total length of the member to decrease by 0.25 mm. The values of elastic modulus for steel and aluminium are 210 GPa and 70 GPa, respectively. (11)

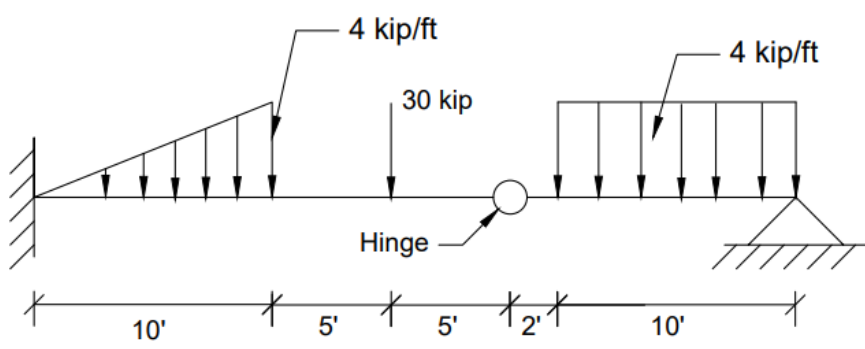


- (c) Determine the largest weight W that can be supported by the two wires shown in the following figure. The stress in either wire is not to exceed 30 ksi. The cross-sectional areas of the wires AB and AC are 0.4 in² and 0.5 in² respectively. (11)

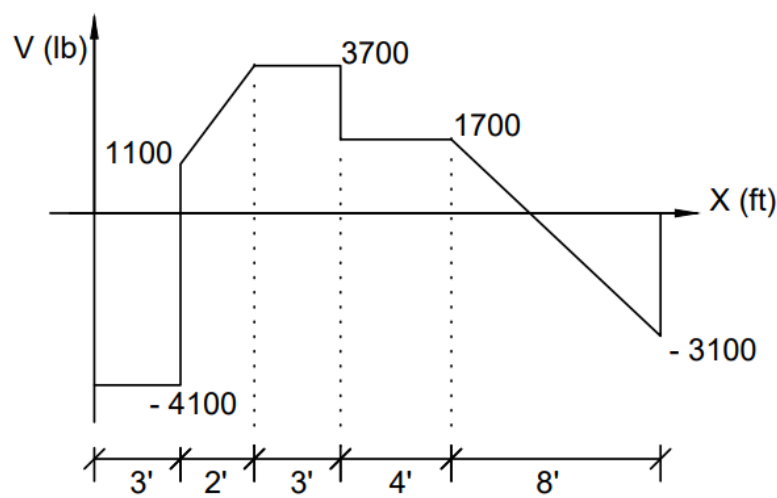


2. (a) Define the terms shear force and bending moment. What do you understand by the term 'point of contraflexure'. (06)
- (b) Draw the 'sign convention' for shear and moment in a beam. (04)

- (c) Draw the shear force and bending moment diagram for the following beam. (20)



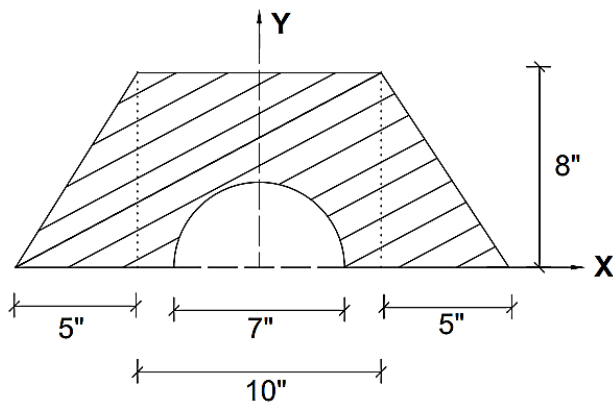
3. (a) Draw moment and load diagram from the following shear diagram. Specify values at all changes of load positions and at all points of zero shear. (15)



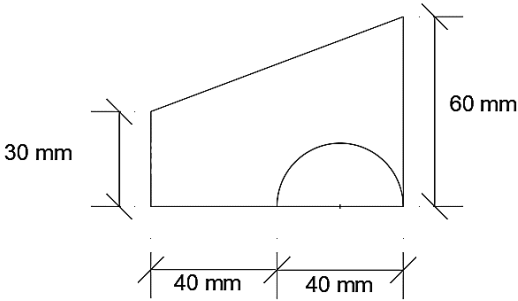
- (b) The ultimate shear stress of a mild steel flat is 500 N/mm². Calculate the diameter of a hole that can be punched through it without exceeding a compressive stress of 250 MPa, in the punch thickness of the flat being 15 mm. Also, determine the strain in the punch when Young's modulus of its material is 2x10⁵ N/mm². (15)

Section – B

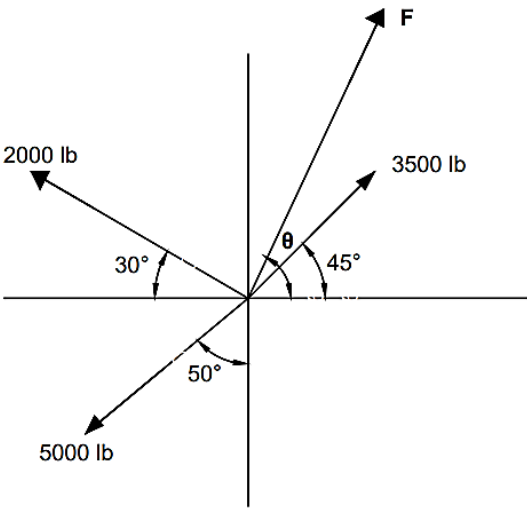
4. (a) What is moment of inertia of area and radius of gyration?. (06)
- (b) For the shaded area of the following figure compute moment of inertia about x-axis and y-axis and also, determine the moment of inertia about a centroidal axis parallel to x and y-axis (24)



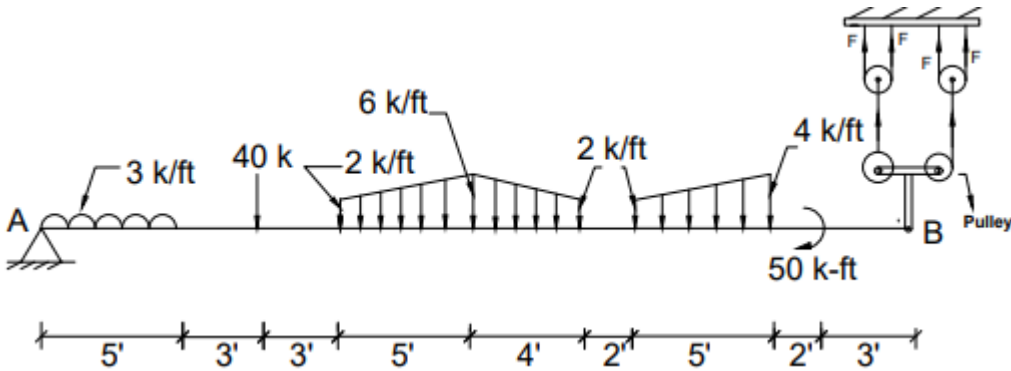
5. (a) Define the terms "centroid" and "centre of gravity". Find the centroid of a "right circular cone" whose altitude is 'h' and radius is 'r'. (14)
- (b) A semicircular area is removed from a trapezium as shown in the following figure. Determine the centroid of the remaining area. (16)



6. (a) What is force? Classify the force system in details. (07)
- (b) Find out the force, F and angle, θ from the following coplanar concurrent force system. (08)



- (c) Find out the value of force, F for the following figure. Assume that all pulleys are frictionless. (15)



Khulna University of Engineering & Technology
 Department of Urban and Regional Planning
 BURP 3rd Year 1st Term Regular Examination, 2020
CE 3151
 (Fundamentals of Transportation Engineering)

Full Marks: 120

Time: 1.5hrs

- N.B.** i) Figures in the right margin indicate full marks.
 ii) Assume suitable data if necessary.
 iii) Answer any two questions from each section in separate script.

Section – A

- | | | |
|---|---|----|
| 1 | (a) What is Highway Engineering? Discuss the phase that dealt by Highway Engineering. | 10 |
| | (b) What are the objectives of land-use planning and control? Draw a typical Cross-section of Urban and Rural Roads. | 10 |
| | (c) What are the objectives of providing traffic separators on roads? Classify kerb according to the height and describe each of them. | 10 |
| 2 | (a) Explain the factors on which the overtaking sight distance depends. Derive an expression for calculating the overtaking sight distance on a road. | 14 |
| | (b) Calculate the extra widening required for a pavement of width 7m on a horizontal curve of radius 250m if the longest wheel base of the vehicle expected on road is 6.1m. Design speed is 80 Kmph. | 06 |
| | (c) Compute the minimum sight distance to avoid head on collision of two buses approaching from the opposite directions. The speed of the buses are 65 Kmph and 80 Kmph, respectively and both bus has brake efficiency of 80%. | 10 |
| 3 | (a) Why are the transition curves introduced at the beginning and end of the sharp horizontal curves? What is valley curve? | 08 |
| | (b) What is “Critical length of gradient” and “Floating gradient”? Describe the design steps for superelevation. | 12 |
| | (c) A vertical summit curve is formed at the intersection of two gradients +1.5% and - 4.0%. Design the length of summit curve to provide a stopping sight distance for a design speed of 90 Kmph. | 10 |

Section – B

- | | | |
|---|---|----|
| 4 | (a) Describe the analyzing Process of the Transportation demand in details. | 12 |
| | (b) What is O-D survey? Write down the common methods of O-D survey. | 08 |
| | (c) What are the objectives of parking survey? Describe off-street parking with suitable examples. | 10 |
| 5 | (a) Define the followings: | 08 |
| | (i) Parking Volume | |
| | (ii) Parking Load | |
| | (iii) Parking Index | |
| | (iv) Parking Turnover | |
| | (b) Write down the process of parallel parking. If there are 350 bays available and 260 bays occupied by the cars, then find out the parking index. | 10 |
| | (c) What are field based survey for public transport? Describe the estimation of goods movement demands. What is PCU? | 12 |
| 6 | (a) What is roadway capacity? Classify it and write down the importance of the concept of highway capacity. | 10 |
| | (b) What are the survey techniques used to collect urban freight data? | 10 |
| | (c) Describe the importance of infrastructure for economic growth. | 10 |
